

Maths Unit 3, Part 3C: Venn word problems and counting formulas

Aasan Notes for Seemab · kitab ke page 57 se 59 tak · practice: Exercise 3.2

1. Application of Venn diagrams (p57, p58)

Kitab ka jumla

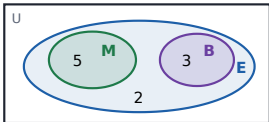
“Venn diagram is a practical mathematical tool for solving real world problems of set theory.”

Lafzon wala sawal aaye to seedha formula mat lagao, pehle **naqsha** banao. Aur naqsha hamesha **andar se bahar** ki taraf bharo: sab se pehle beech wala hissa (jo teeno mein aata hai), phir do-do ke hisse, phir akele wale, aur aakhir mein rectangle ke bahar ka number. Jo log do cheezon mein aate hain woh donon circles mein gine ja chuke hote hain, is liye seedha jorne se jawab hamesha zyada aata hai.

Example 11 (p57), subset wala case

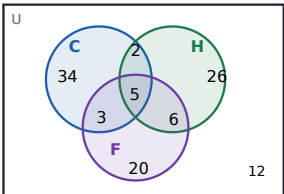
“Among the ten teachers of a secondary school, five teach Mathematics and three teach Biology. However, all these teachers also teach English. Show the data by Venn diagram. Also find how many teachers teach only English.”

Solution. E = set of English Teachers, M = set of Mathematics Teachers, B = set of Biology Teachers. *Both M and B are the subsets of E*, so dono chhote ovals bare oval E ke **andar** banenge. Only English = $10 - 5 - 3 = \mathbf{2 \text{ teachers}}$.



Example 12 (p57, p58), teen overlapping sets padhna

“In a survey, people were asked whether they like cricket, hockey or football.” Naqshe se jawab nikalne hain. C = cricket, H = hockey, F = football.



Number of people who play	From the figure	Notation
(i) only cricket	34	$n(C - C \cap H \cap F)$
(ii) hockey and football	$6 + 5 = 11$	$n(H \cap F)$
(iii) all three games	5	$n(C \cap H \cap F)$
(iv) either of the three games	$34 + 5 + 6 + 26 + 20 = 91$	$n(C \cup H \cup F)$
(v) neither of the three games	12	$n(U - C \cup H \cup F)$
Total number of people surveyed	$34 + 5 + 6 + 26 + 20 + 12 = 103$	$n(U)$

Kitab ki ghalti Example 12 mein, exam mein yeh galti mat karna

Kitab (iv) mein sirf $34 + 5 + 6 + 26 + 20 = 91$ jorti hai aur **2** (cricket aur hockey wale) aur **3** (cricket aur football wale) ko bhool jati hai. **Saray saat hisse joro** to $n(C \cup H \cup F) = 34 + 2 + 3 + 5 + 6 + 26 + 20 = 96$, aur $n(U) = 96 + 12 = 108$. Naqshe mein bhi **3** C aur F ke jore wale hisse se thora bahar chhap gaya hai; uski asal jagah wohi lens hai. Doosri baat: “only cricket” ka sahi notation **$n(C) - n(C \cap H) - n(C \cap F) + n(C \cap H \cap F)$** hai, kitab wala $n(C - C \cap H \cap F)$ nahin.



2. Application of set theory: counting formulas (p58)

Ab naqsha banaye baghair ginti. **n(A)** ka matlab hai set A ke elements ki tadaad. Union ke formula mein ghatana isi liye parta hai ke jo dono sets mein hain woh **do dafa** gin liye gaye, ek dafa $n(A)$ mein aur ek dafa $n(B)$ mein.

Case	Formulae (kitab p58)
(i) For any two overlapping sets A and B	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$ $n(A - B) = n(A \cup B) - n(B)$ $n(A - B) = n(A) - n(A \cap B)$ $n(A \cup B) = n(A - B) + n(A \cap B) + n(B - A)$
(ii) For any two sets A and B that are disjoint	$n(A \cup B) = n(A) + n(B)$ $n(A - B) = n(A)$
(iii) For any three sets A, B and C	$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

★ Key Fact (p58), yahan se MCQ banti hai

For any sets A and B: $A \cup A = A$, $A \cap A = A$, $A \cap \phi = \phi$, $A \cup \phi = A$, $A \cap B \subseteq A$, $A \subseteq A \cup B$.

✏ Example 13 (p58, p59), do sets ka formula

"In a class of 80 students, 40 like English, 34 like Mathematics and 9 like both. How many students like either of both subjects and how many like neither?"

Total number of students = $n(T) = 80$ · $n(E) = 40$ · $n(M) = 34$ · $n(E \cap M) = 9$

$n(E \cup M) = n(E) + n(M) - n(E \cap M) = 40 + 34 - 9 = 65$ (like either subject)

$n(T) - n(E \cup M) = 80 - 65 = 15$ (like neither subject)

⚠ Example 13 ke lafzon par dhyan

Kitab likhti hai ~~total number of students that like both subjects is $n(E \cup M) = 65$~~ . Yeh lafz ghalat hain. 65 woh hain jo **either** subject pasand karte hain (kam az kam ek). **Both** to sirf 9 hain, yaani $n(E \cap M)$. Isi tarah 15 woh hain jo **neither** subject pasand karte hain.

🧠 Pakki trick

Sawal ke lafz seedha formula batate hain:
both ya **and** = $n \cdot \text{either, or, at least one} = n(A \cup B)$
U · **neither, none** = $n(U) - n(A \cup B)$ · **only A** = $n(A) - n(A \cap B)$. Sirf teen sets wale formula mein signs ka order yaad rakho: **plus, plus, plus, minus, minus, minus, plus**. Ab **Exercise 3.2** khud karo.

📖 Quiz

Ye teeno sawal tumhare Maths paper mein bhi aayenge, is liye inhein achhi tarah karo.

- Write the formulae.** (i) $n(A \cup B \cup C)$ for any three sets A, B and C. (ii) Both formulae the book gives for $n(A - B)$ when A and B are overlapping.
- Solve.** In a survey of 120 students, 60 play cricket, 45 play hockey, 30 play football, 20 play cricket and hockey, 15 play hockey and football, 12 play cricket and football, and 8 play all three games. Find (i) $n(C \cup H \cup F)$, (ii) how many play none of the three games, (iii) how many play only football.
- Explain in one or two lines.** In Example 13 we write $n(E \cup M) = n(E) + n(M) - n(E \cap M)$. Why must $n(E \cap M)$ be subtracted instead of added?